

Support Vector Machine

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Outline

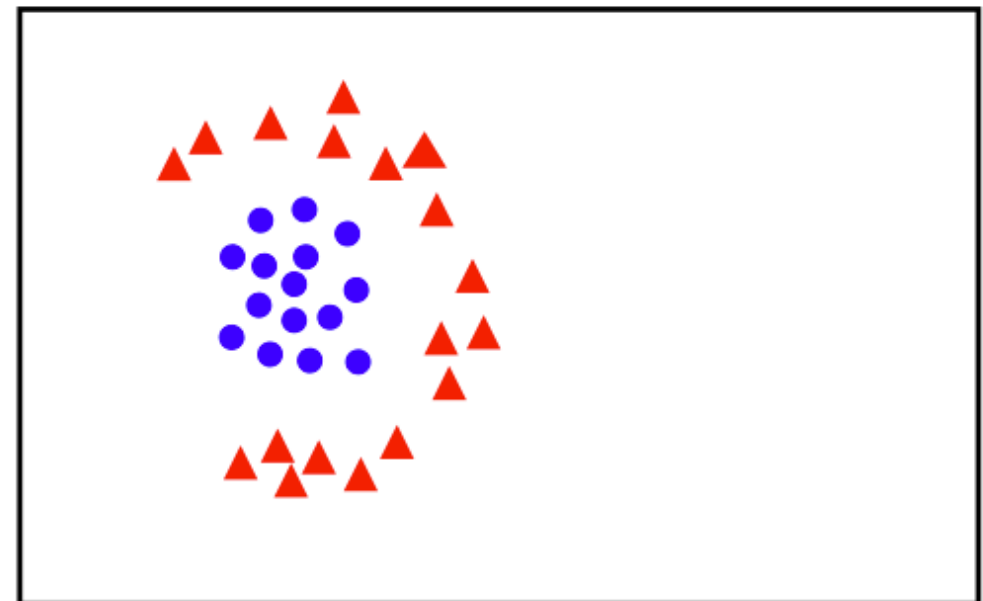
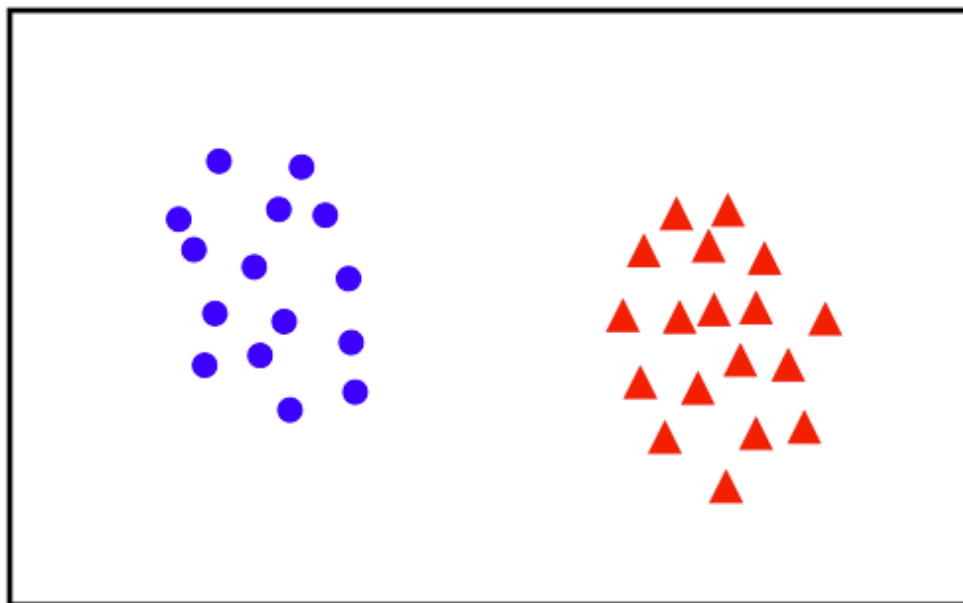
- Precursor: Linear Classifier and Perceptron ←
- Support Vector Machine
- Parameter Learning

Binary Classification

Given training data $(x^{\{i\}}, y^{\{i\}})$ for $i = 1 \dots N$, with $x^{\{i\}} \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

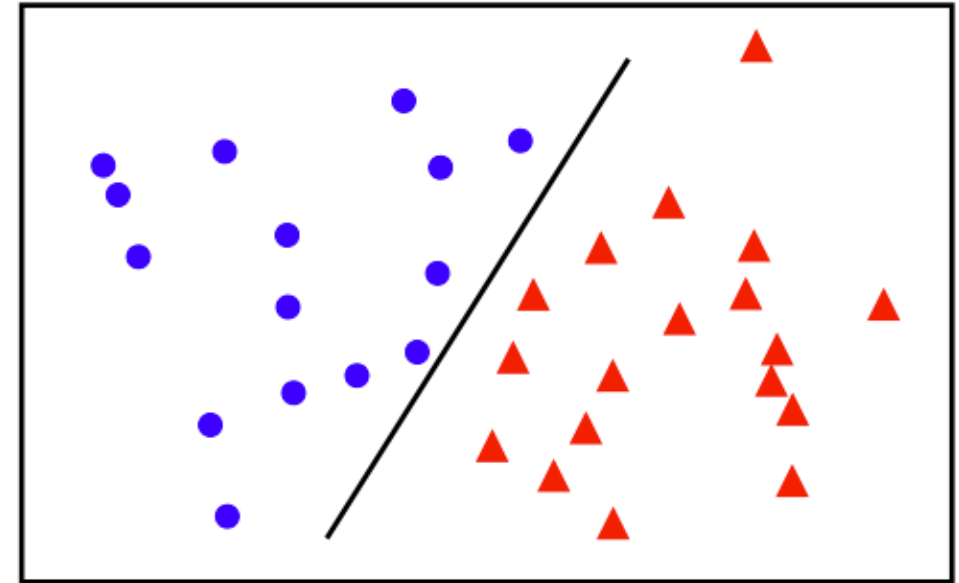
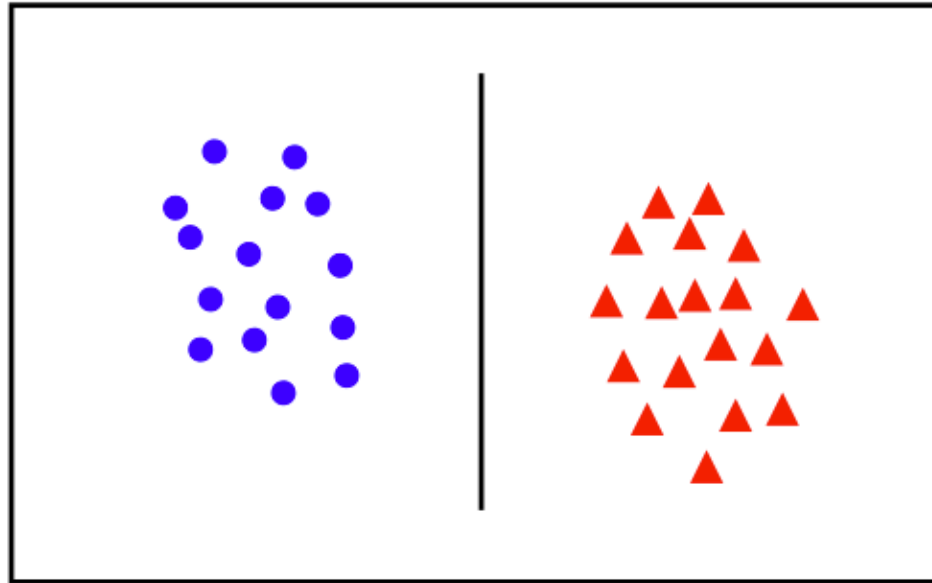
$$f(x^{\{i\}}) \begin{cases} \geq 0 & +1 \\ < 0 & -1 \end{cases}$$

i.e. $y^{\{i\}} f(x^{\{i\}}) > 0$ for a correct classification.

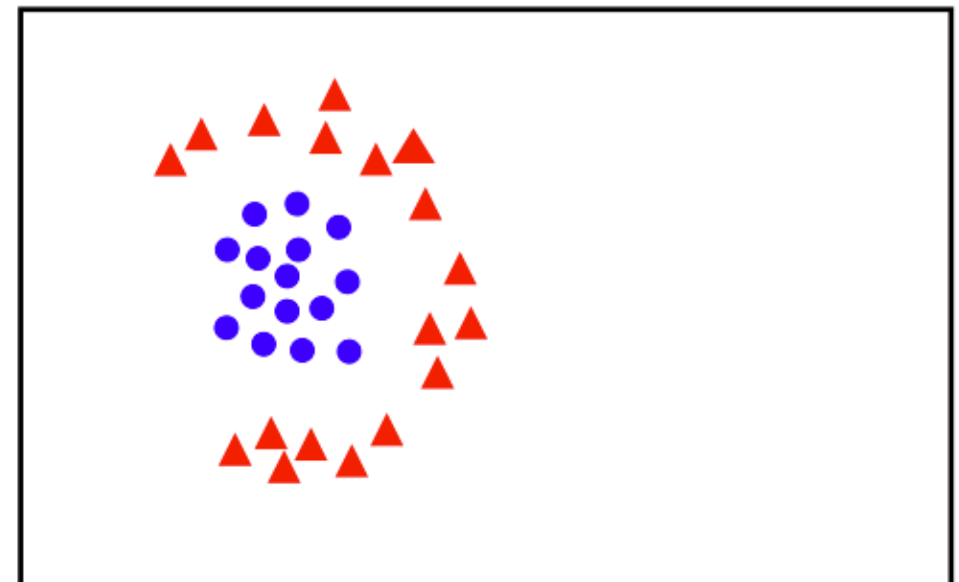
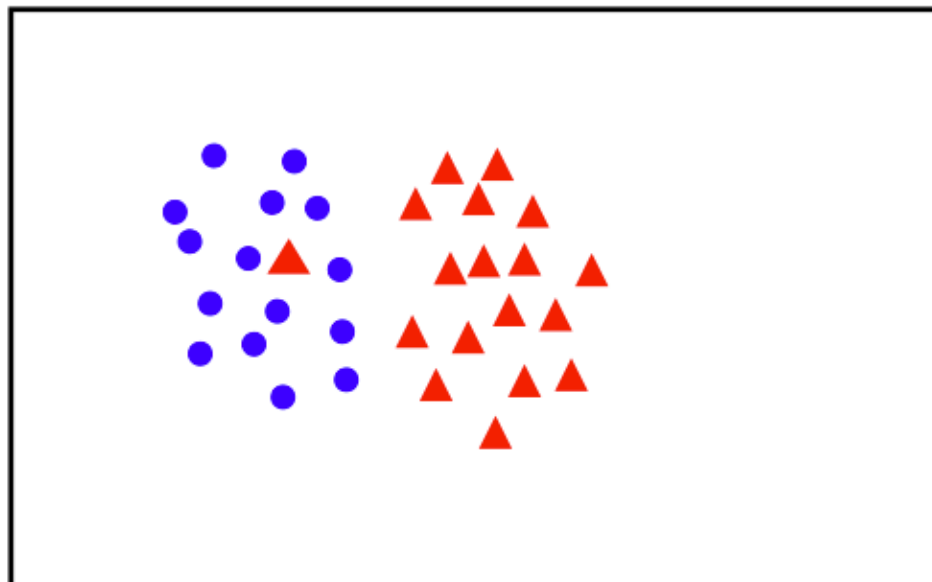


Linear Separability

linearly
separable



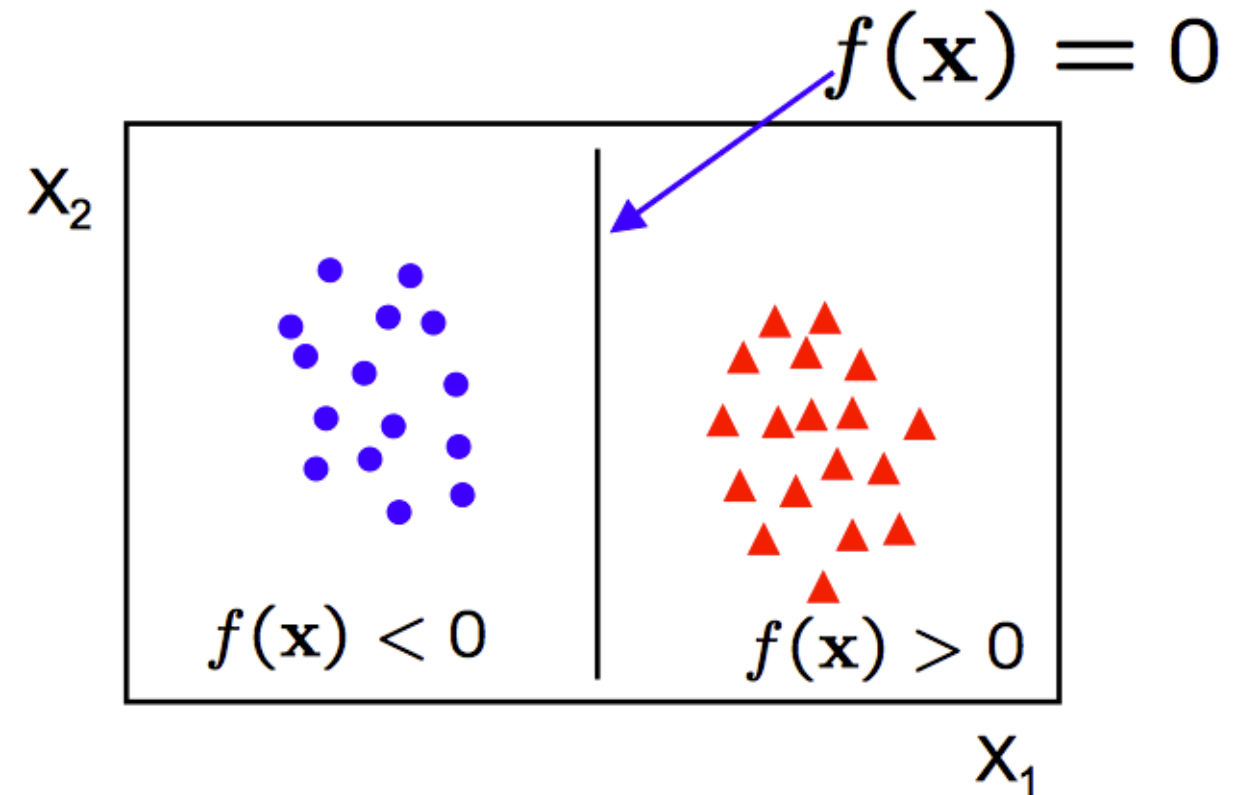
not
linearly
separable



Linear Classifier

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$

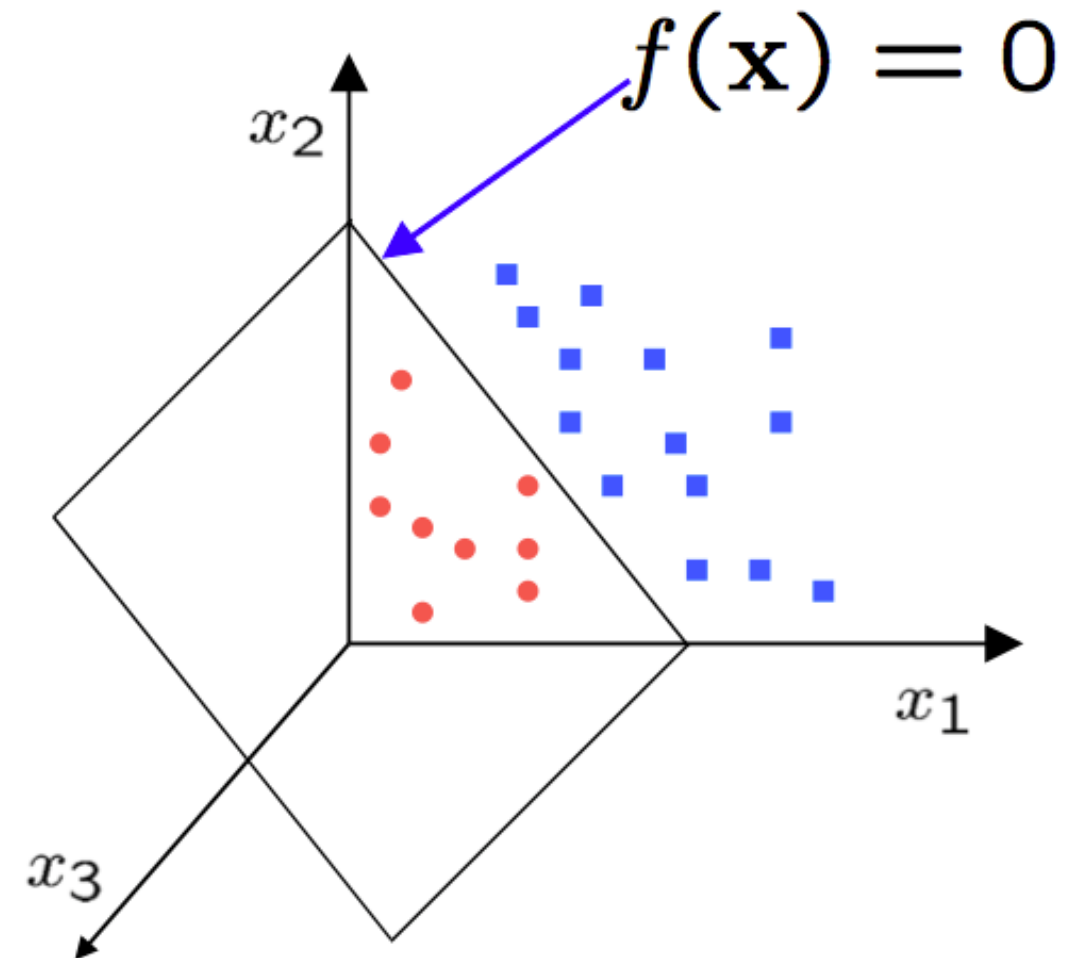


- in 2D the discriminant is a line
- θ is the **normal** to the line, and θ_0 the **bias**
- θ is known as the **weight vector**

Linear Classifier (higher dimension)

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$



- in 3D the discriminant is a plane, and in nD it is a hyperplane

The Perceptron Classifier

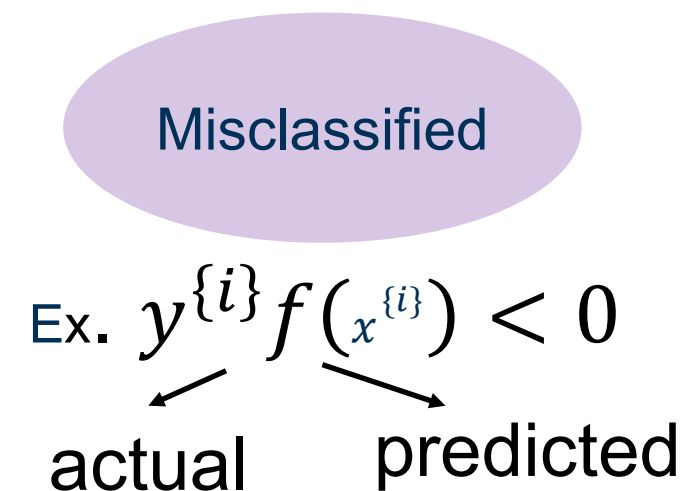
Considering $\mathbf{x}$ is linearly separable and $\mathbf{y}$ has two labels of $\{-1, 1\}$

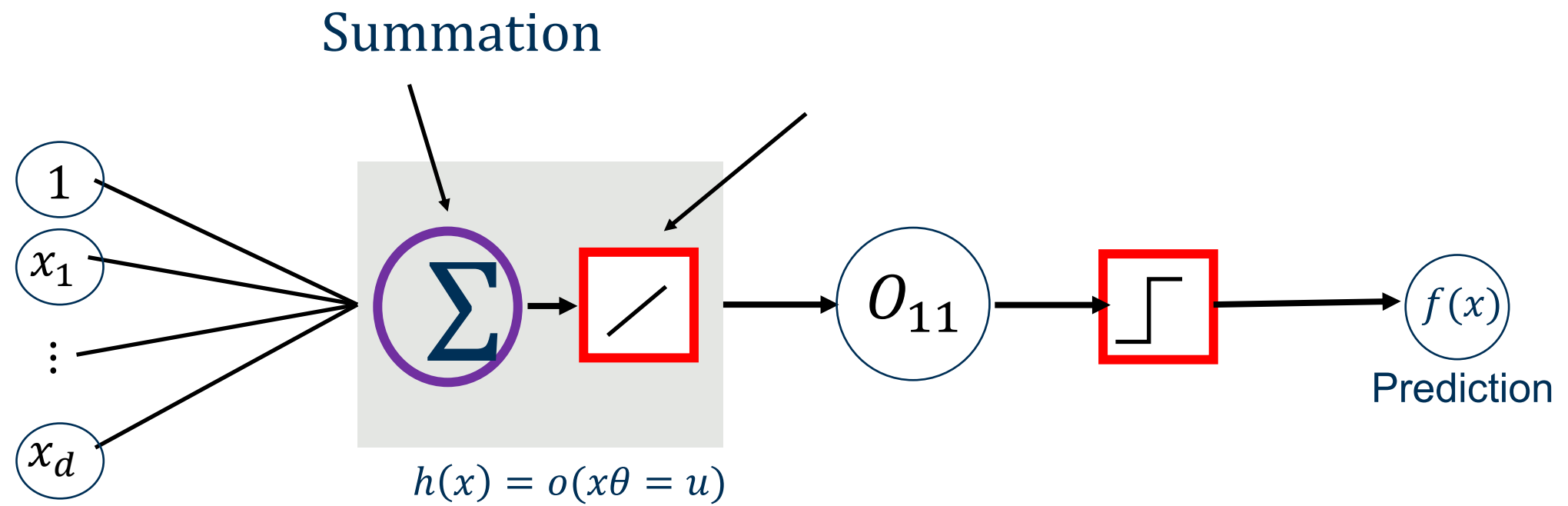
$$f(x^{(i)}) = x^{(i)} \theta \quad \text{Bias is inside } \theta \text{ now}$$

How can we separate datapoints with label 1 from datapoints with label -1 using a line?

Perceptron Algorithm:

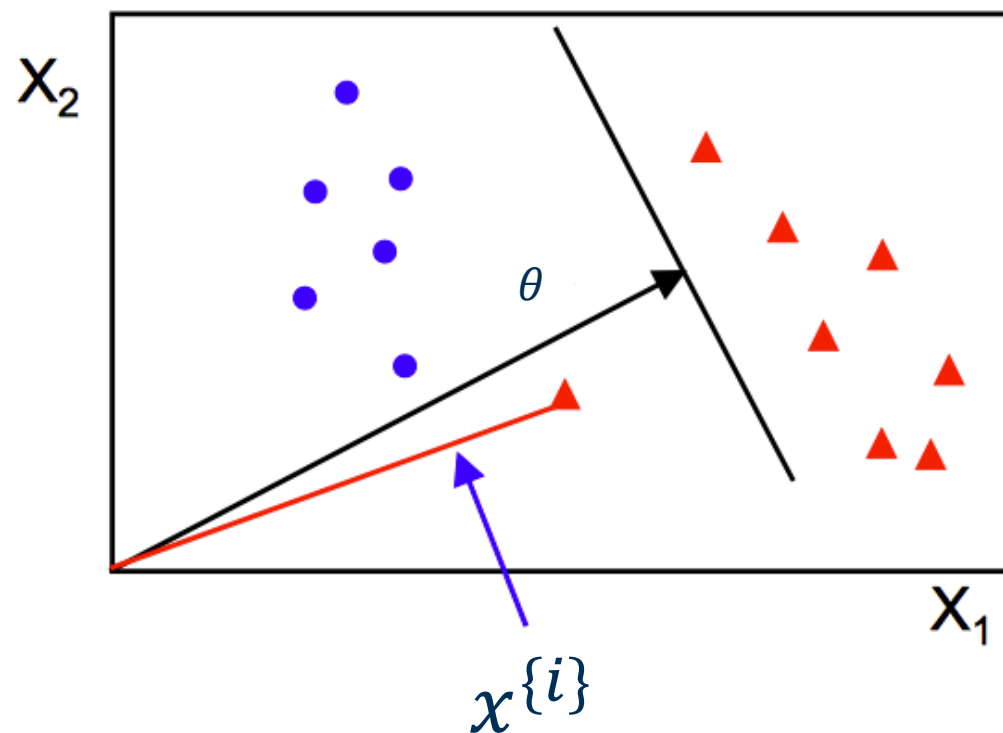
- Initialize $\theta = 0$
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
- If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified



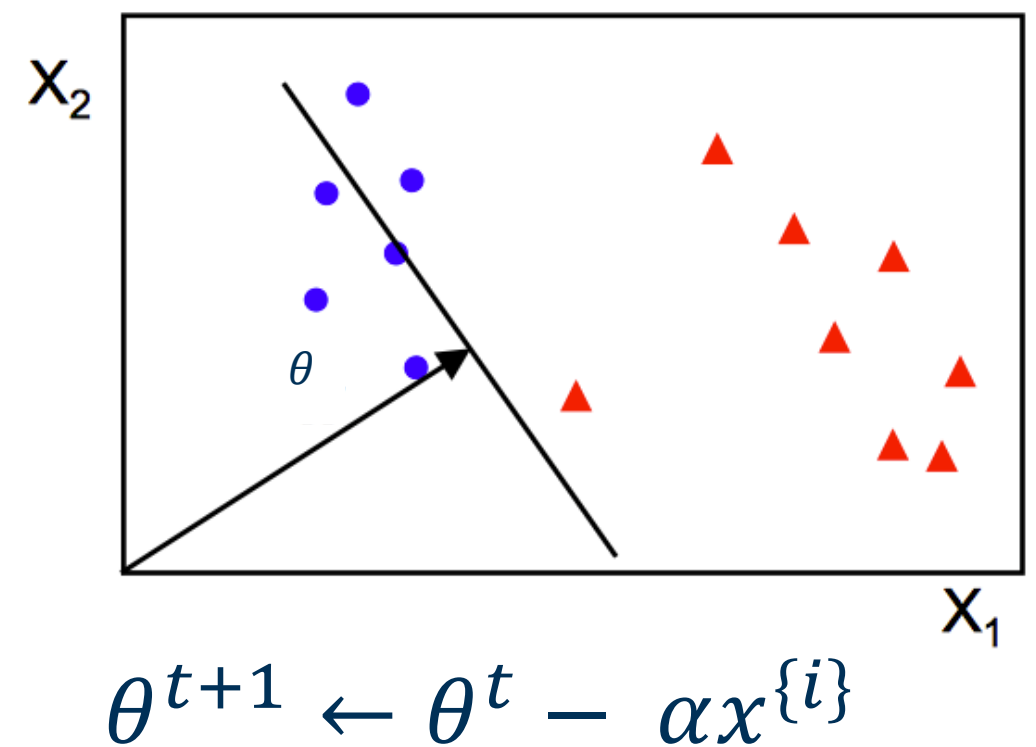


- Initialize $\theta = 0$
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
- If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified

before update



after update

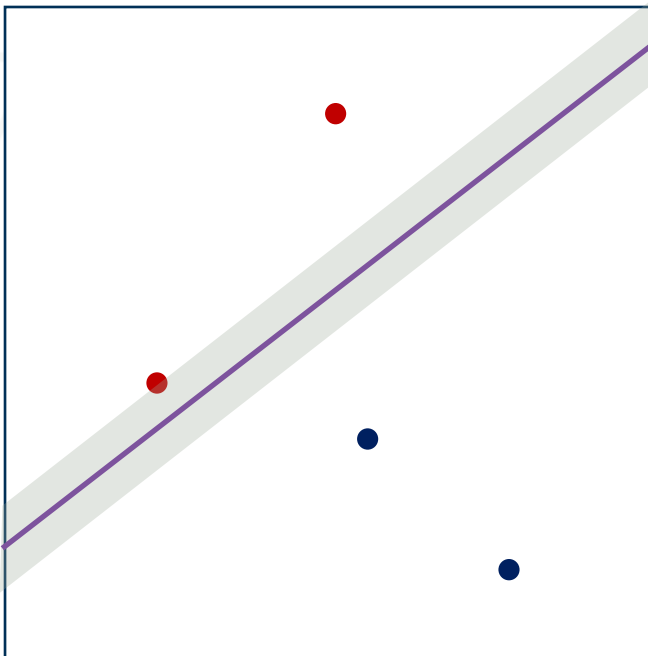


$$\theta^{t+1} \leftarrow \theta^t - \alpha x^{(i)}$$

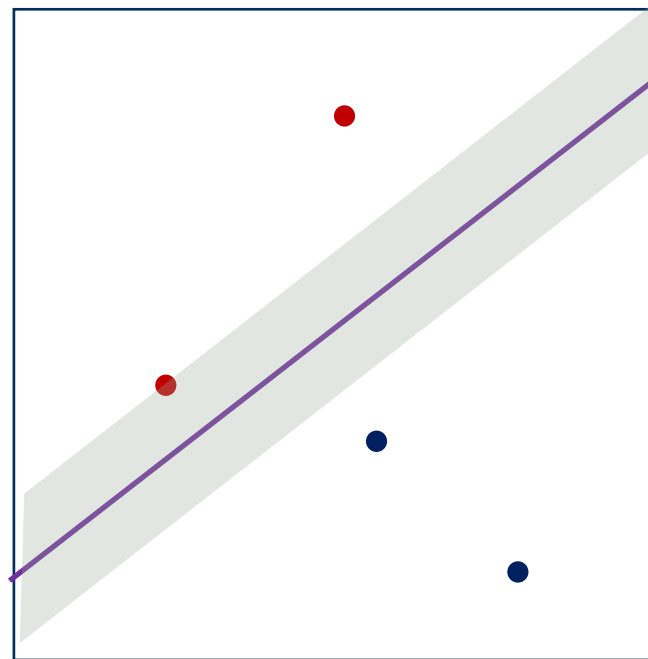
Linear separation

We can have different separating lines

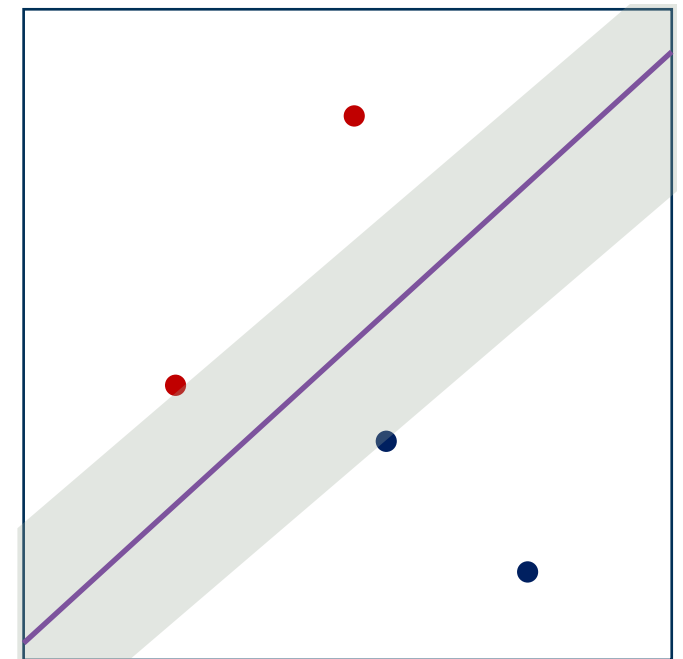
1



2



3



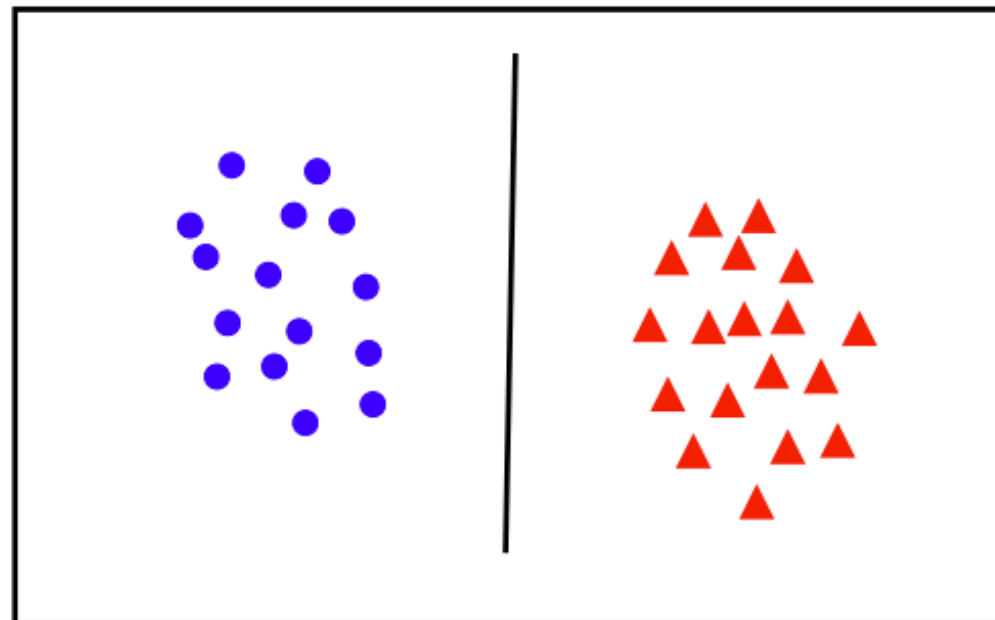
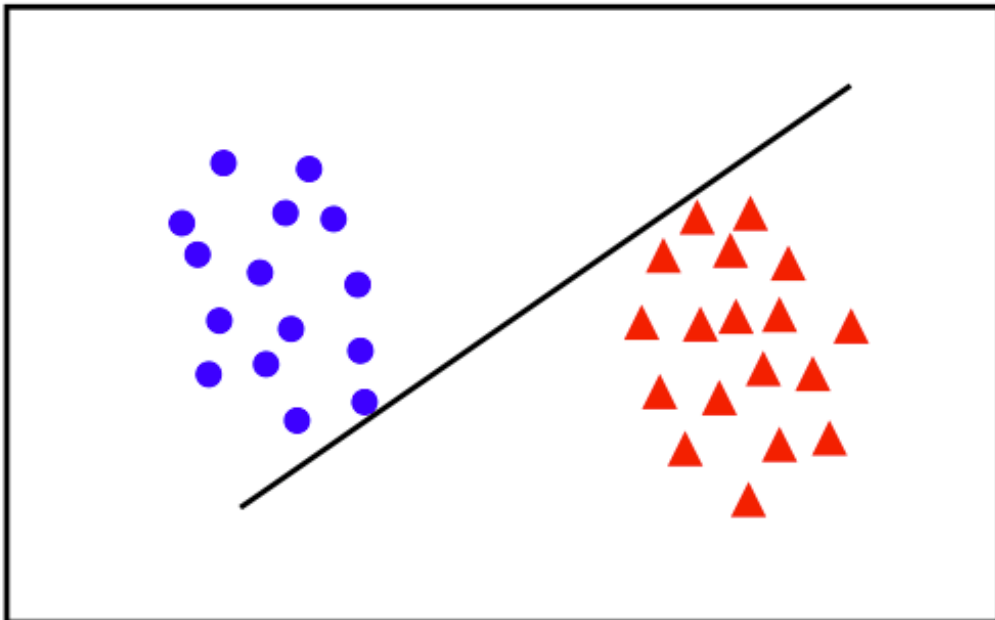
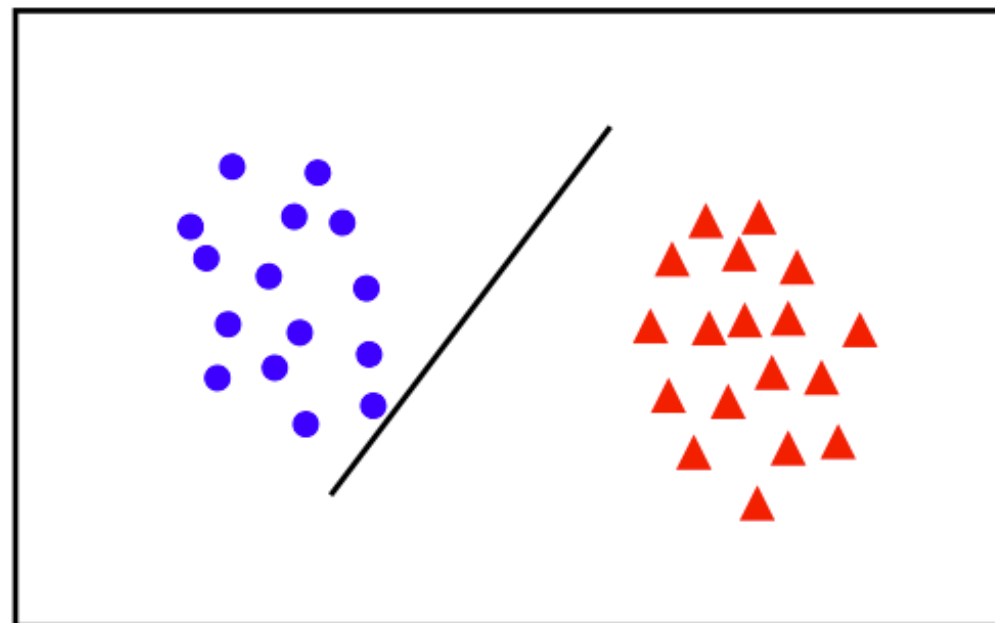
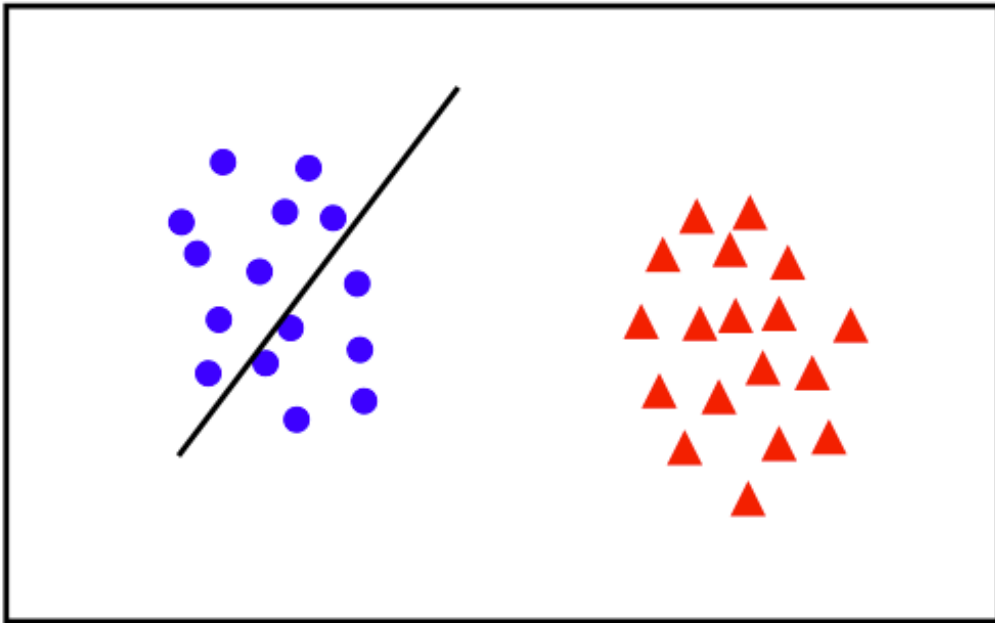
Which line is the best?

All cases, error is zero and they are linear, so they are all good for generalization.

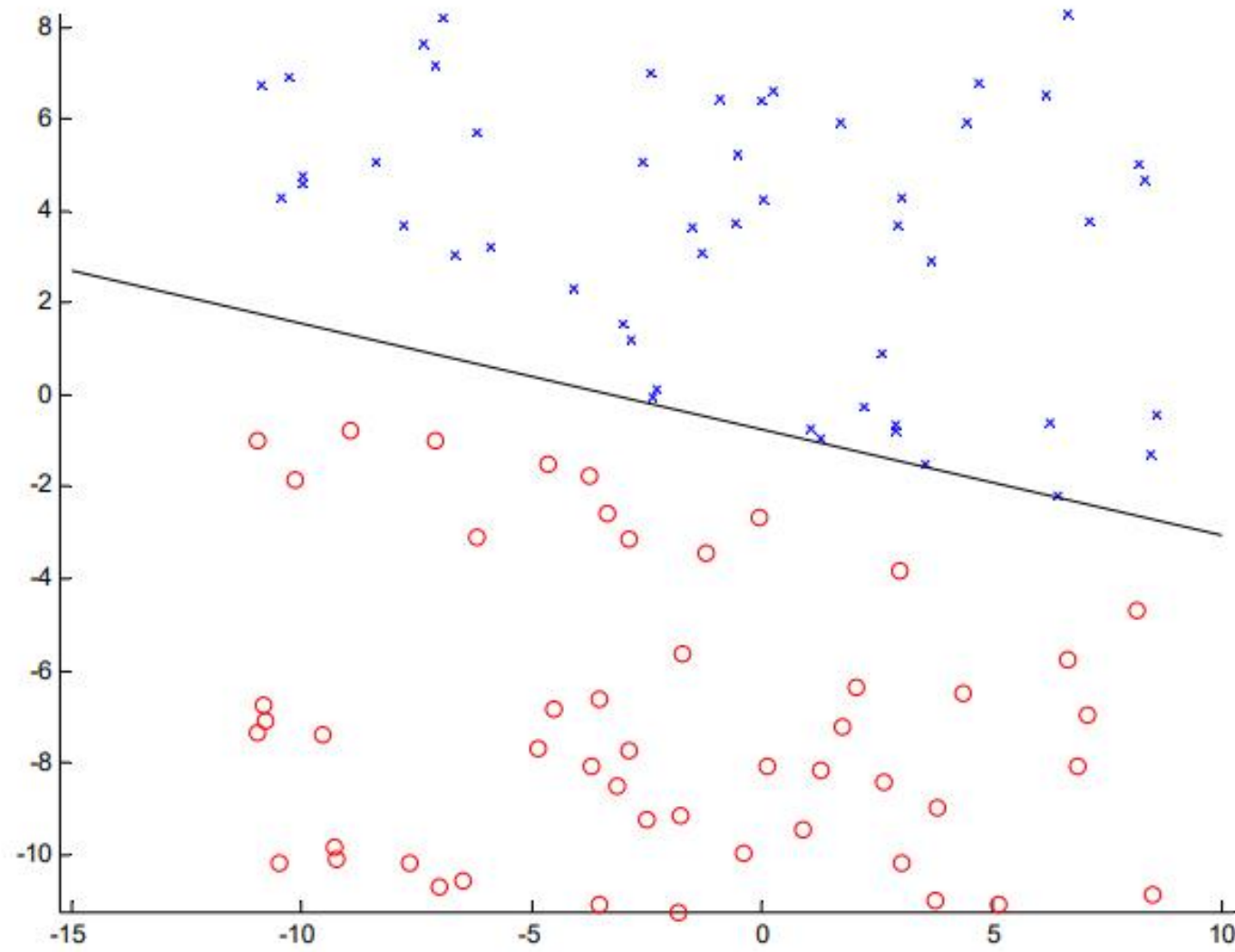
Why is the bigger margin better?

What θ maximizes the margin?

What is the Best θ ?



Perceptron example



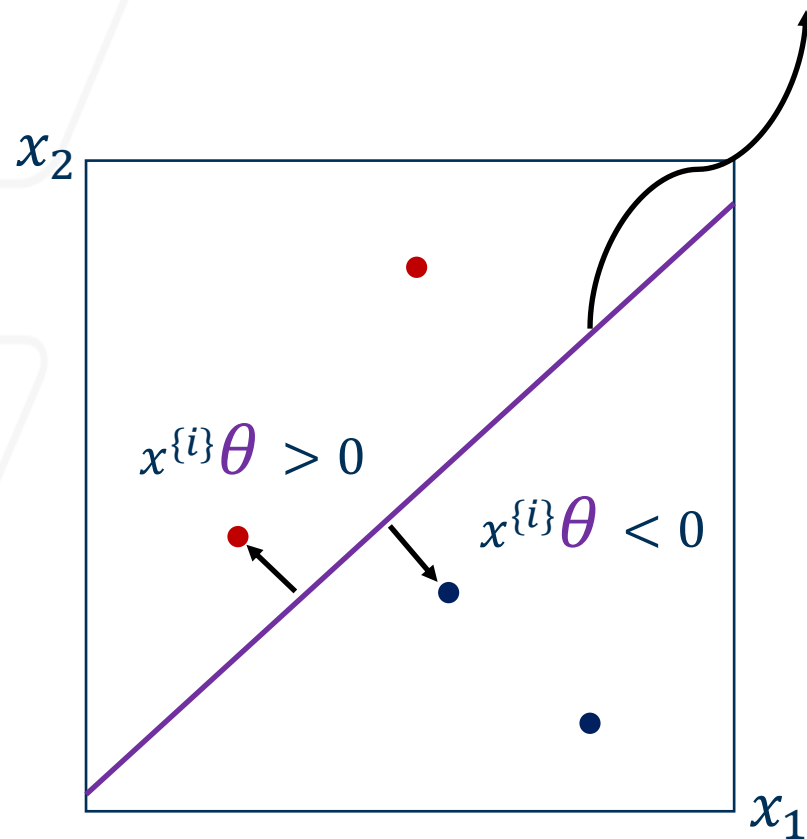
- if the data is linearly separable, then the algorithm will converge
- convergence can be slow ...
- separating line close to training data
- we would prefer a larger margin for generalization (better generalization)

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Finding θ with a fat margin

Solution (decision boundary) of the line: $x\theta = 0$

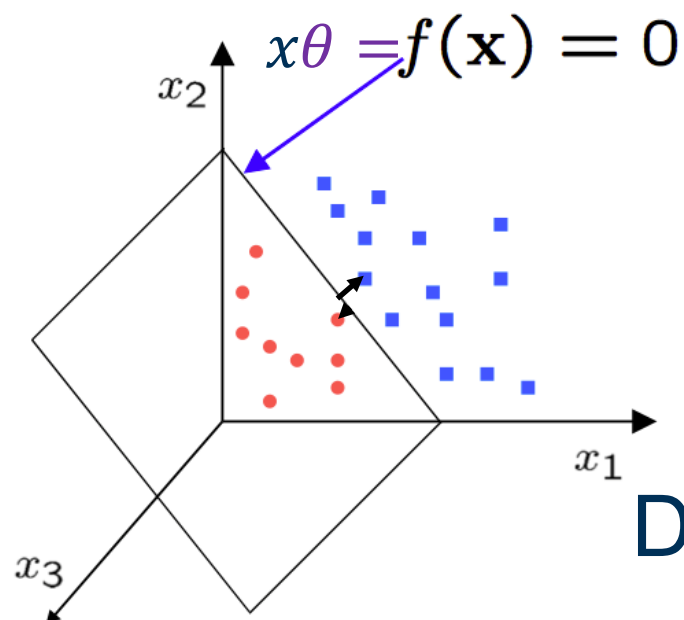


Let $x^{(i)}$ to be the nearest data point to the line (plane):

$$|x^{(i)}\theta| > 0$$

Our line solution is $x\theta = 0$

Does it matter if I scale up or down θ for the decision boundary?



$$|x^{(i)}\theta| = 1 \rightarrow \text{normalization}$$

Let's pull out θ_0 from $\theta = (\theta_1, \dots, \theta_d)$ and call it be b

Decision boundary would be: $x\theta + b = 0$

Computing the distance

The distance between $x^{\{i\}}$ and the line $x\theta + b = 0$ where $|x^{\{i\}}\theta + b| = 1$

The vector θ is perpendicular to the decision line.

You should ask me why?

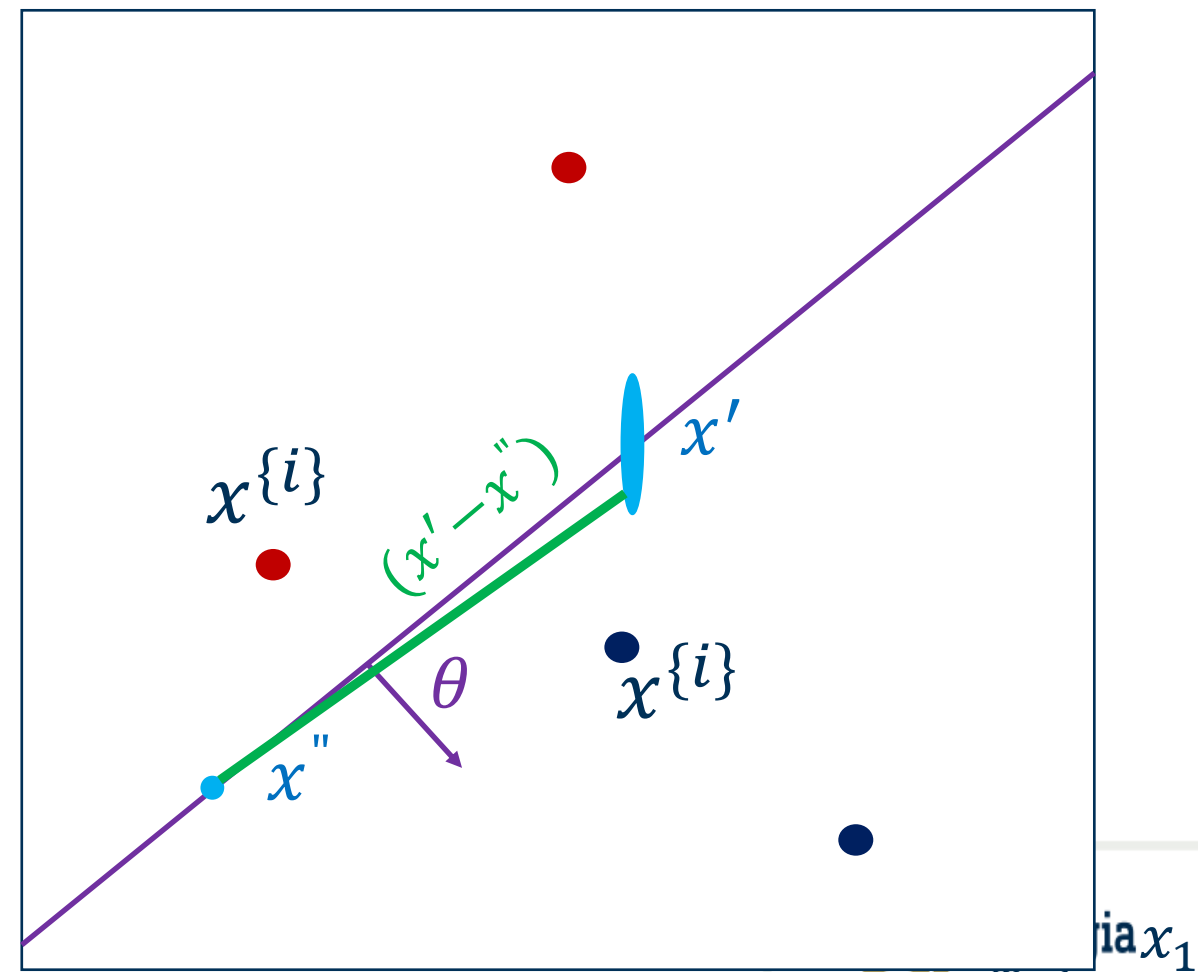
Consider x' and x'' on the plane

$$x'\theta + b = 0 \quad \text{and} \quad x''\theta + b = 0$$

$$\Downarrow$$
$$x'\theta + b = x''\theta + b$$

$$\Rightarrow (x' - x'')\theta = 0$$

x_2



ia x_1

What is the distance of my fat margin?

What is the distance between $x^{\{i\}}$ and the plane?

Let's take any point x on the line:

Distance would be projection of $(x^{\{i\}} - x)$ vector on θ .

To project the vector, we need to normalize θ to get the unit vector.

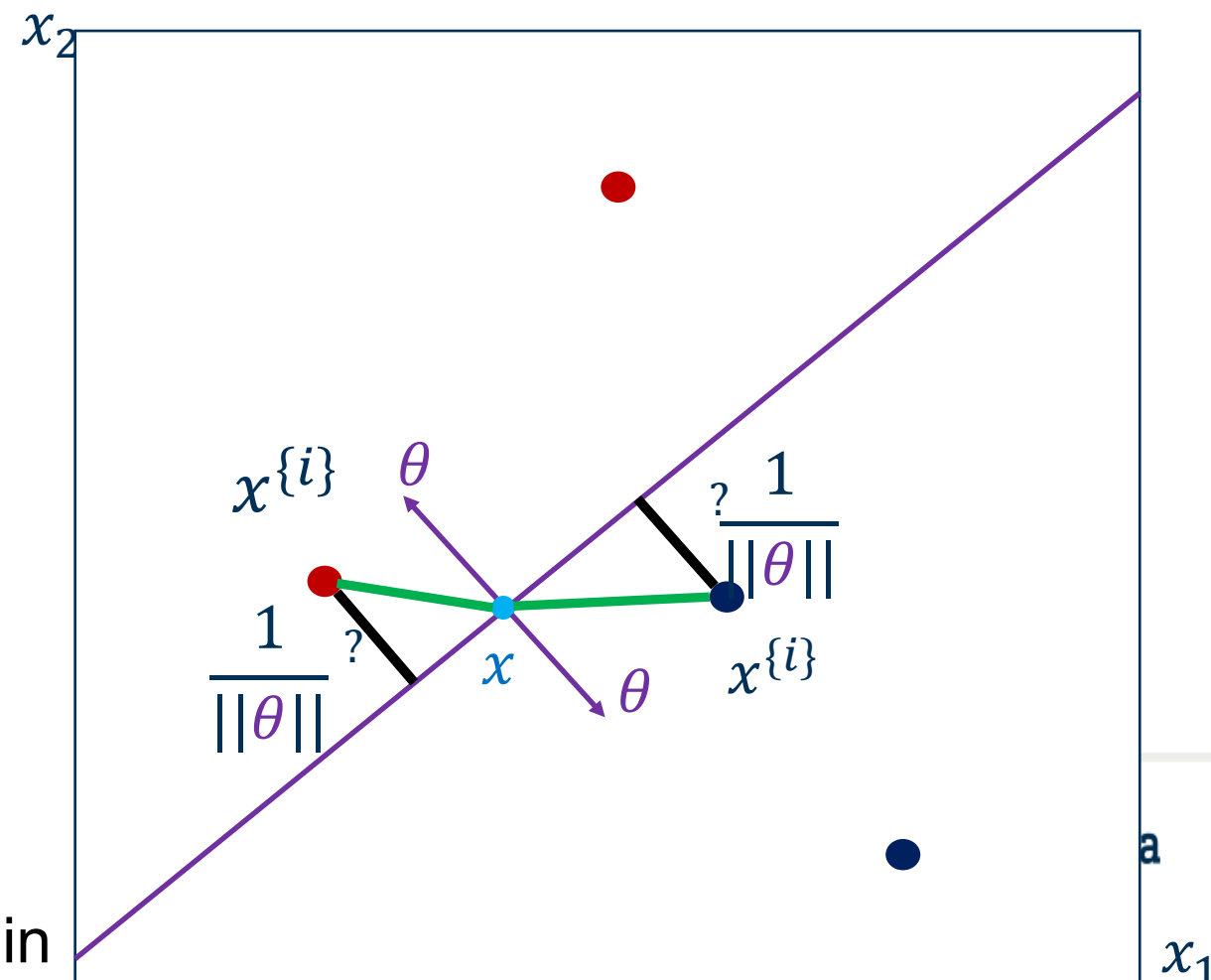
$\hat{\theta} = \frac{\theta}{\|\theta\|} \Rightarrow \text{distance} = |(x^{\{i\}} - x)\hat{\theta}|$ which is the dot product

$$\text{distance} = \frac{1}{\|\theta\|} |(x^{\{i\}}\theta - x\theta)|$$

$$= \frac{1}{\|\theta\|} |(\underbrace{x^{\{i\}}\theta + b}_{\text{My constraint } |x^{\{i\}}\theta + b| = 1} - \underbrace{x\theta + b}_{\text{A point on the decision line } x\theta + b = 0})|$$

$$= \frac{1}{\|\theta\|}$$

The margin



Now we need to maximize the margin

Maximize

$$\frac{1}{||\theta||}$$

Subject to Min value of $|x^{\{i\}}\theta + b| = 1 \Rightarrow \text{nearest neighbour}$
 $i = 1, 2, \dots, N$

There is a “min” in our constraining; it can be hard to optimize this problem(non-convex form)

Can I write the following term to get rid of absolute value?

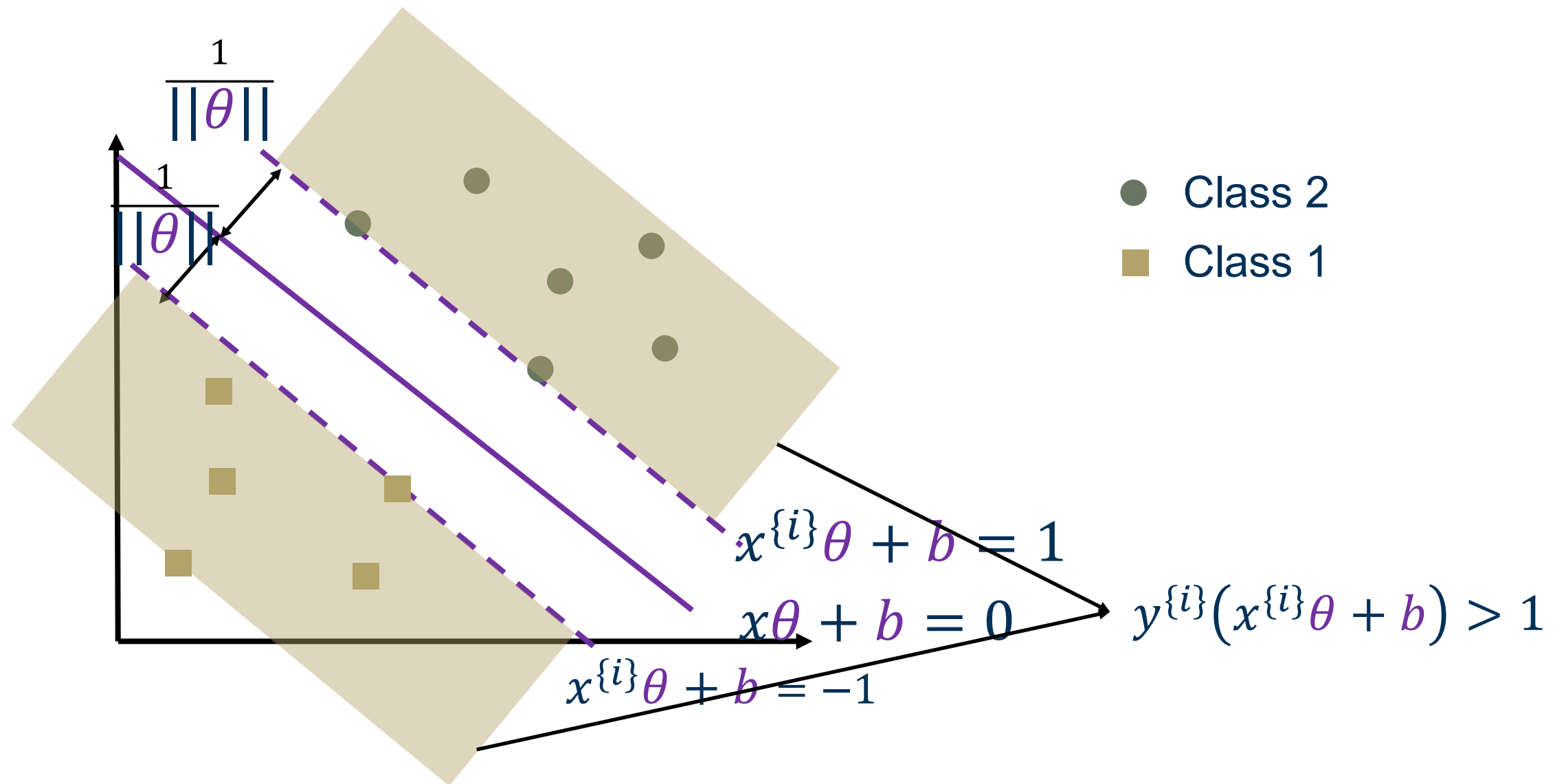
$$|x^{\{i\}}\theta + b| = y^{\{i\}}(x^{\{i\}}\theta + b) \Rightarrow \text{for a correct classification}$$

If min $|x^{\{i\}}\theta + b| = 1 \Rightarrow \text{so it can be at least 1}$

Maximize

$$\frac{1}{||\theta||}$$

Subject to $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$



Maximize $\frac{2}{\|\theta\|}$

Subject to $y^{i}(x^{i}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

If $\theta \neq 0$, there exists a max value

Minimize $\frac{1}{2}\theta\theta^T$

Subject to $y^{i}(x^{i}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

Constrained optimization

$$\text{Minimize } \frac{1}{2} \theta \theta^T$$

$$\text{Subject to } y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1 \text{ for } i = 1, 2, \dots, N$$

$$\theta \in \mathbb{R}^d, b \in \mathbb{R}$$

Using Lagrange method:

But wait, there is an **inequality** in our constraints

We use Karush-Kuhn-Tucker (KKT) condition to deal with this problem

$$g(x) = y^{\{i\}}(x^{\{i\}}\theta + b) - 1 \quad \alpha = \text{lagrange multiplier}$$

We need to optimize

θ, b , and α

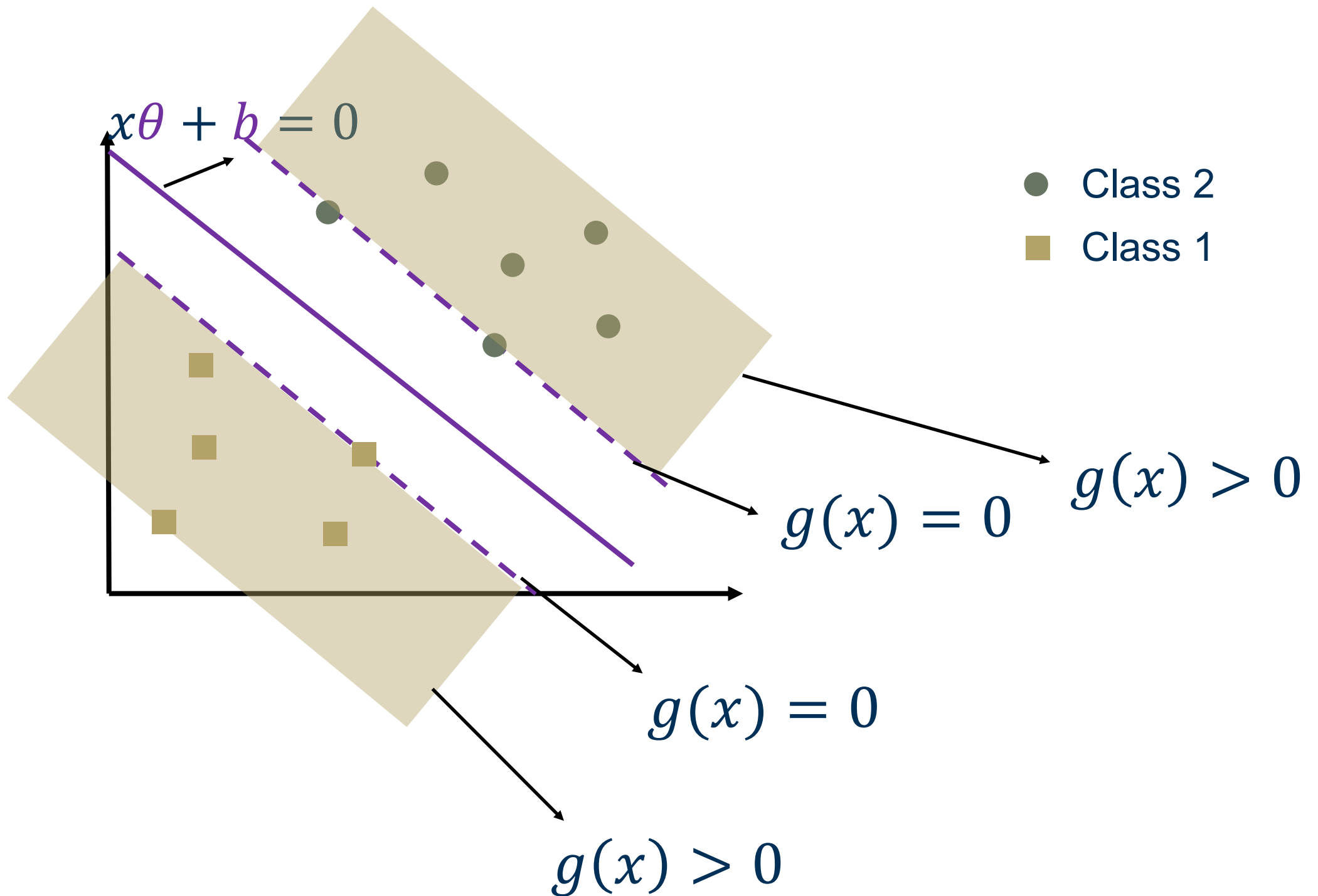
1) Stationary

2) $g(x) \geq 0$ Primal feasibility

3) $\alpha \geq 0$ Dual feasibility

4) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$



$$g(x) = y^{\{i\}}(x^{\{i\}}\theta + b) - 1$$

3) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$

Lagrange formulation

$$\text{Minimize} \quad \frac{1}{2} \theta \theta^T \quad \text{s.t.} \quad y^{\{i\}} (x^{\{i\}} \theta + b) - 1 \geq 0$$

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{\{i\}} (x^{\{i\}} \theta + b) - 1)$$

Minimize w.r.t θ and b and maximize w.r.t each $\alpha_i \geq 0$

$$\nabla_{\theta} \mathcal{L}(\theta, b, \alpha) = \theta - \sum_{i=1}^N \alpha_i y^{\{i\}} x^{\{i\}} = 0$$

$$\nabla_b \mathcal{L}(\theta, b, \alpha) = - \sum_{i=1}^N \alpha_i y^{\{i\}} = 0$$

$$\theta = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$$

$$\sum_{i=1}^N \alpha_i y^{(i)} = 0$$

Let's substitute these in the Lagrangian:

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta + b) - 1) \quad \times$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta + b)) \quad \times$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta)) \quad \uparrow$$

$$= \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \theta \theta^T = \sum_{i=1}^N \alpha_i - \frac{1}{2} \theta \theta^T$$

$$\theta = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$$

$$\sum_{i=1}^N \alpha_i y^{(i)} = 0$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \theta \theta^T$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j x^{(i)} x^{(j)T}$$

maximize w.r.t each $\alpha_i \geq 0$ for $i = 1, \dots, N$

and

$$\sum_{i=1}^N \alpha_i y^{(i)} = 0$$

The solution – quadratic programming

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}T}$$

Quadratic programming packages usually use “min”

$$\min_{\alpha} \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}T} - \sum_{i=1}^N \alpha_i$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}T} & \dots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}T} & \dots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}T} \end{bmatrix} \alpha + (-I^T) \alpha$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}T} & \dots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}T} & \dots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}T} \end{bmatrix} \underbrace{\alpha + (-I^T) \alpha}_{\text{Linear term}}$$

Quadratic coefficients

Subject to

$$\sum_{i=1}^N \underbrace{\alpha_i y^{\{i\}}}_{\text{Linear equality constraint}} = y^T \alpha = 0$$

Linear equality constraint

Pass these to a quadratic programming package

$$\text{lower bound}(0) \leq \alpha \leq \text{upper bound}(\infty)$$

$$\min_{\alpha} \frac{1}{2} \alpha^T Q \alpha - 1^T \alpha \quad \text{subject to} \quad y^T \alpha = 0; \alpha \geq 0$$

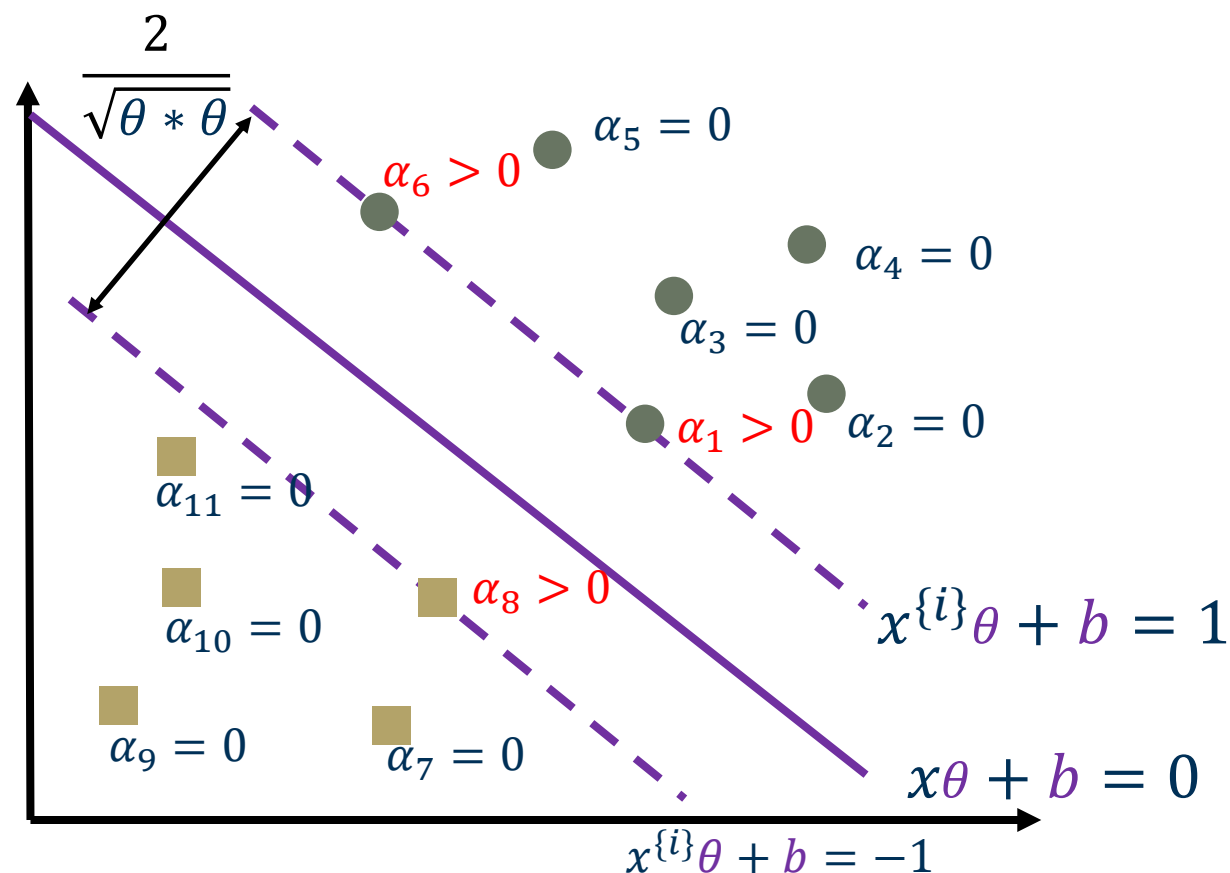
Quadratic programming will give us α

Solution: $\alpha = \alpha_1, \dots, \alpha_N$

KKT condition ($\alpha_i g_i(\theta) = 0$): $\alpha_i (y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) > 0 \quad \Rightarrow \quad \alpha_i = 0$$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0 \quad \Rightarrow \quad \alpha_i > 0 \Rightarrow x_i \text{ is a support vector}$$



Training

$$\theta = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$$

No need to go over all datapoints

$$\rightarrow \theta = \sum_{x^{(i)} \text{ in } SV} \alpha_i y^{(i)} x^{(i)}$$

and for b pick any support vector and calculate:

$$y^{(i)}(x^{(i)}\theta + b) = 1$$

Testing

For a new test point s

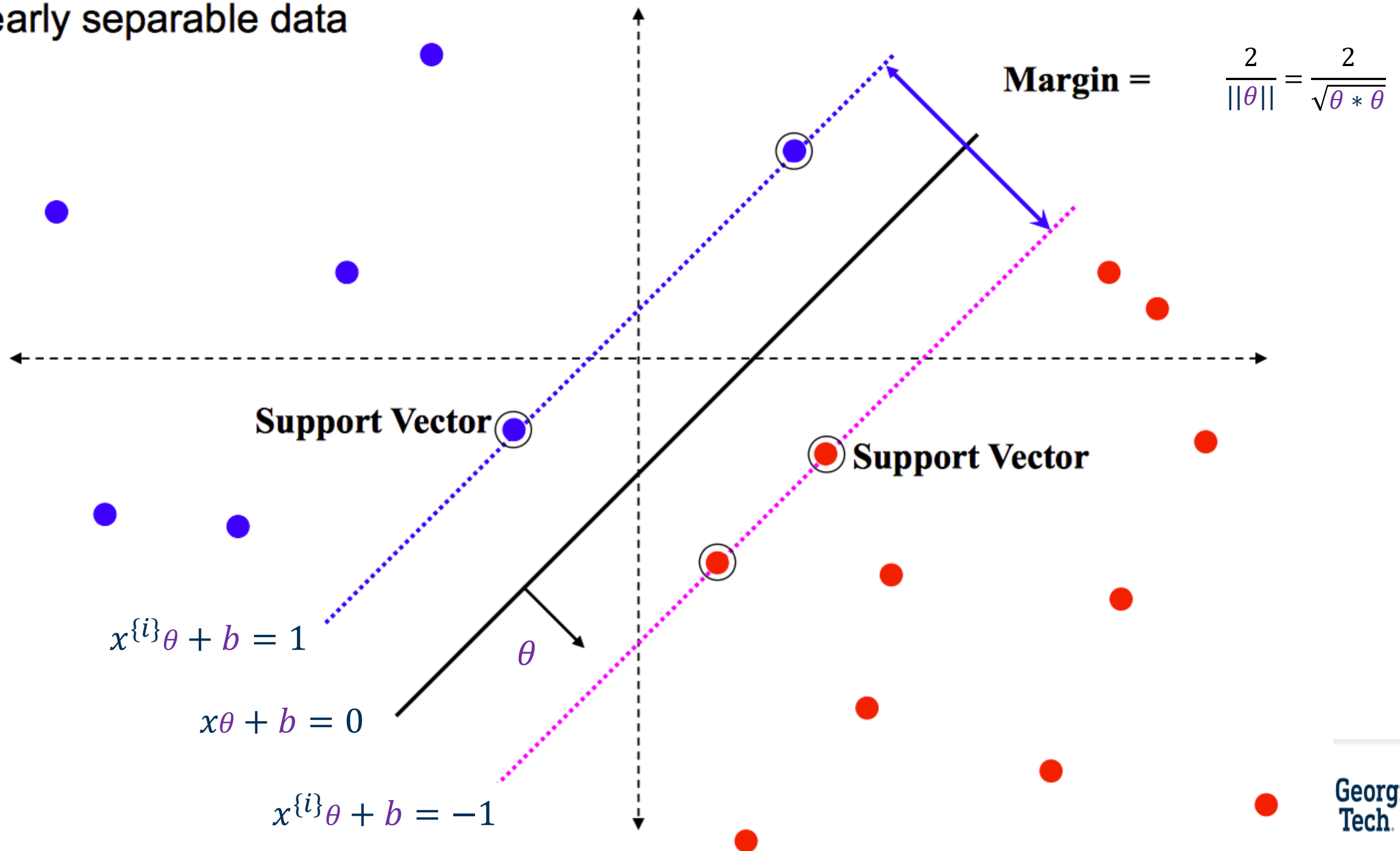
Compute:

$$s\theta + b = \sum_{x^{(i)} \text{ in } SV} \alpha_i y^{(i)} x^{(i)} s^T + b$$

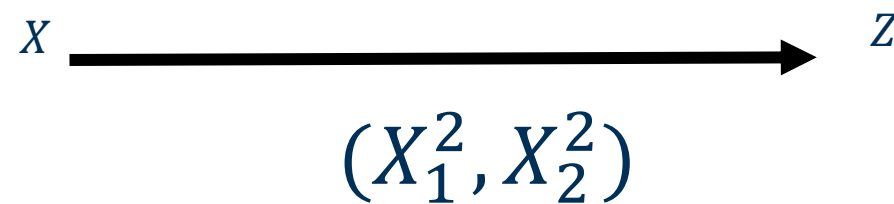
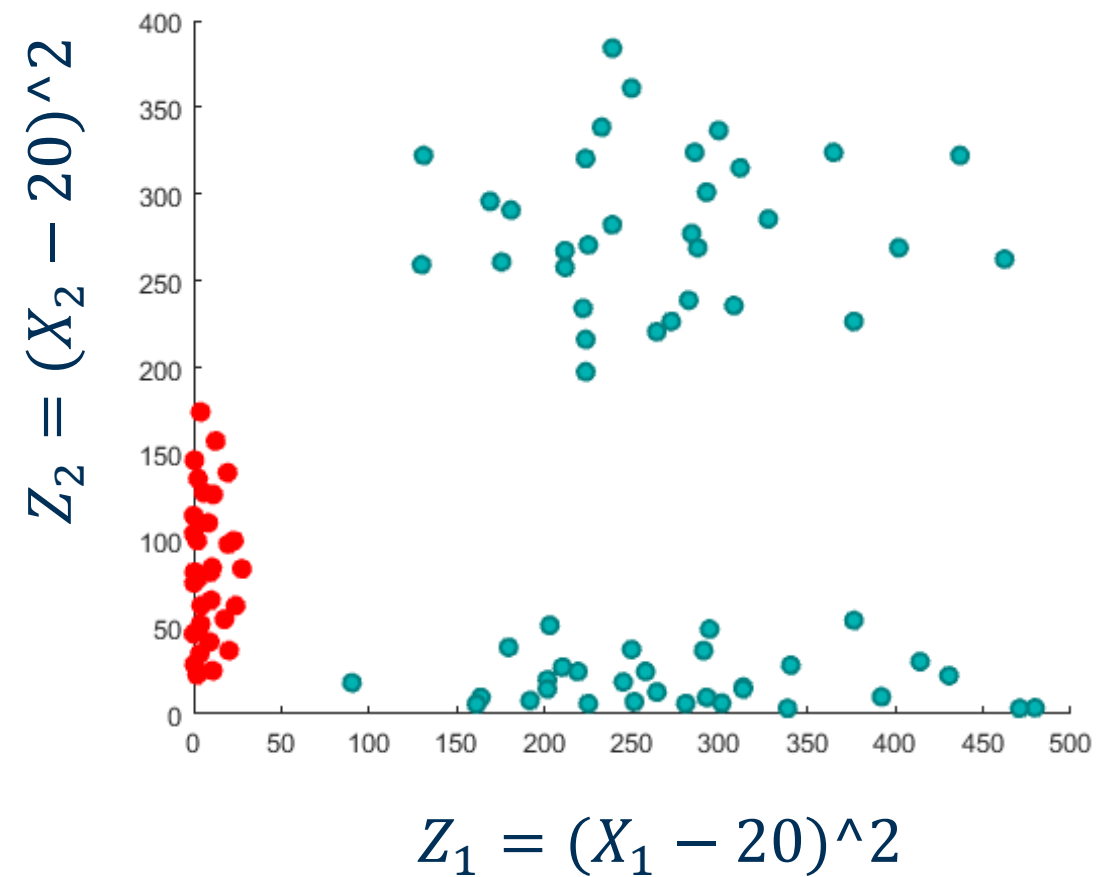
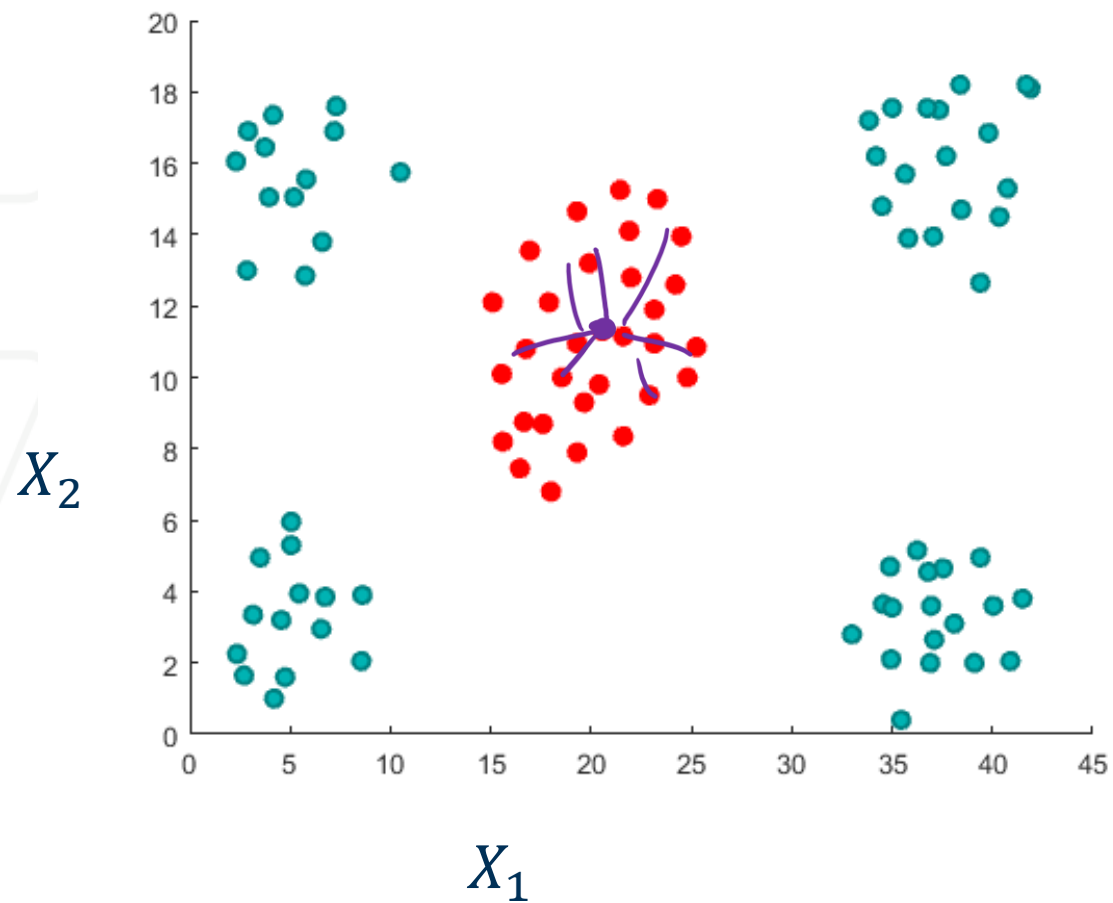
Classify s as class 1 if the result is positive, and class 2 otherwise

Geometric Interpretation

linearly separable data

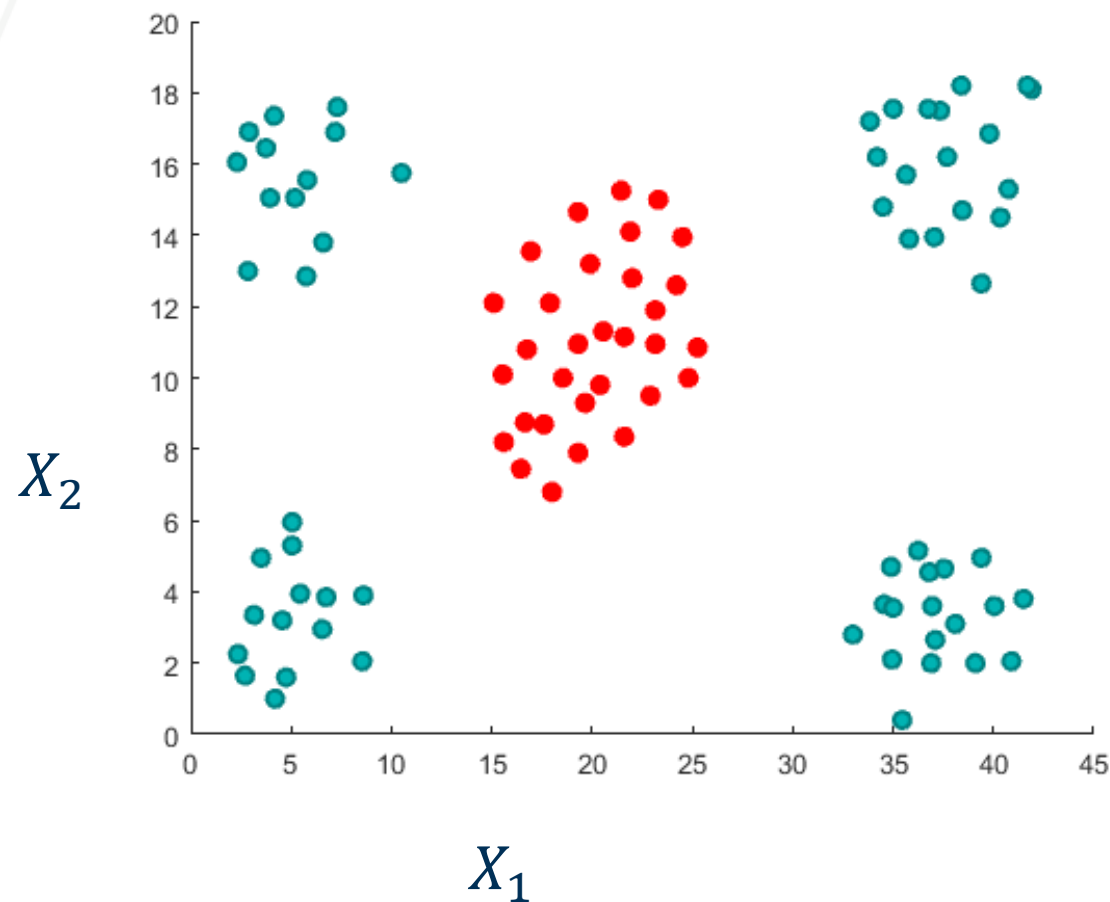


From x to z space



In x space

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j x^{(i)} x^{(j)T}$$

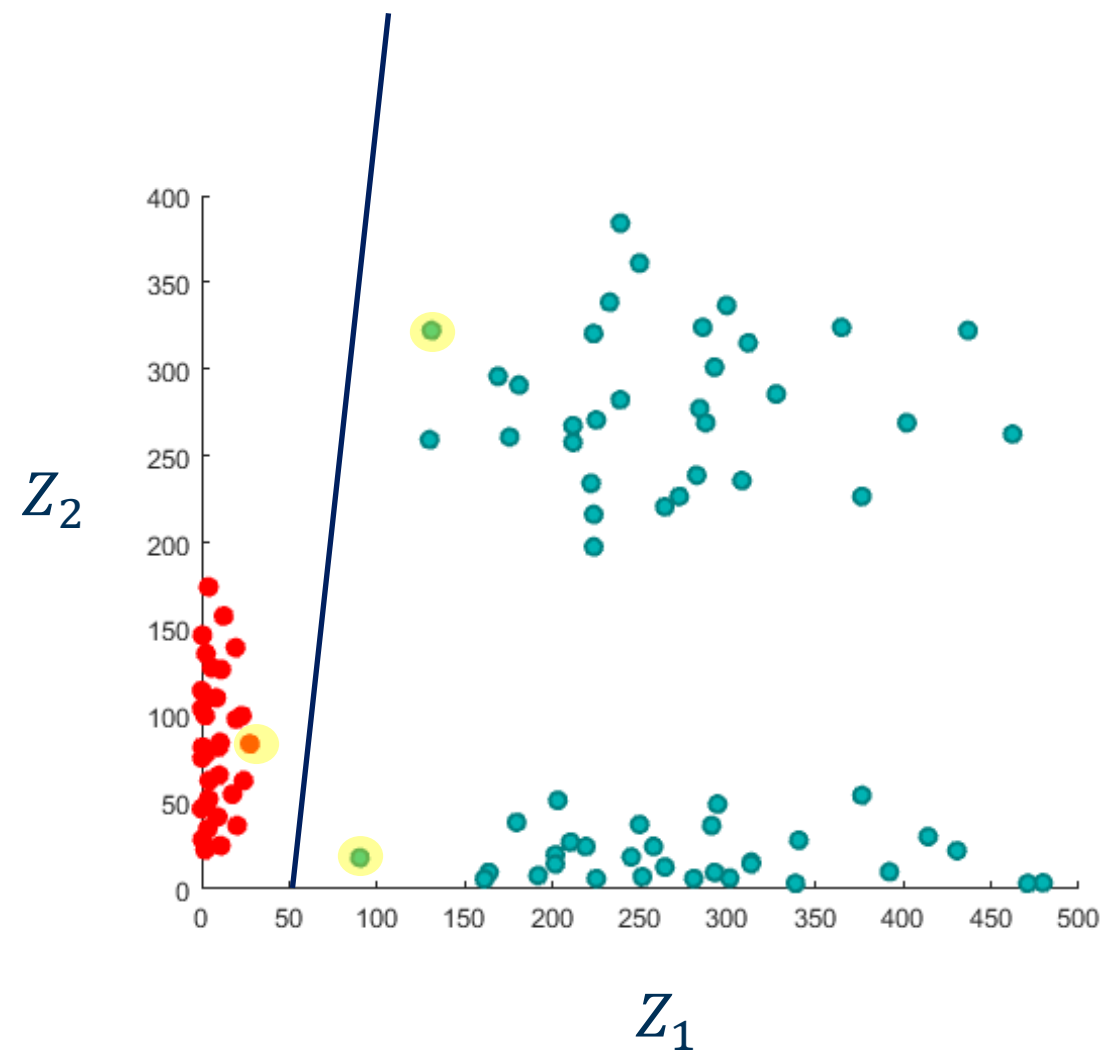


let's say x is $n \times d$
 xx^T will be $n \times n$

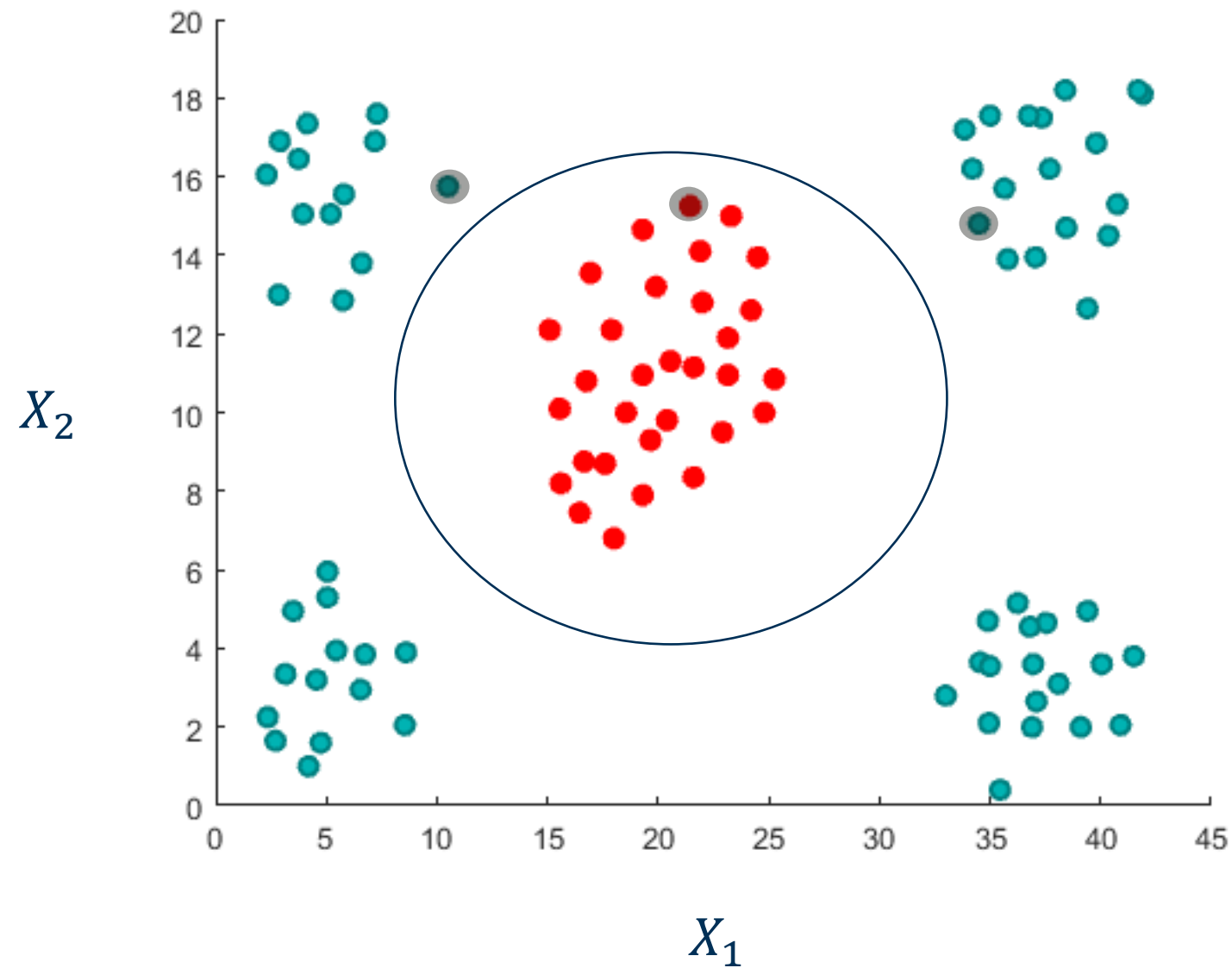
If I add millions of
dimensions to x ,
would it affect the
final size of xx^T ?

In z space

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j \mathbf{z}^{(i)} \mathbf{z}^{(j)T}$$



In x space, they are called pre-images of support vectors



Take-Home Messages

- Linear Separability
- Perceptron
- SVM: Geometric Intuition and Formulation

Quick Knowledge Check